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Subject: Programming with Java	**Semester: III**
ASSIGNMENT No : 11	**Year: AY 2026-27**
TITLE : Interface Implementation	

Problem Statement

Write a Java program to demonstrate the implementation of interfaces.

Learning Objectives

To implement interfaces for achieving abstraction and multiple inheritance of type.

Program:

InterfaceStudentEmployee.java

```
// Assignment 11 - Exercise 1: Printable Interface
import java.util.*;
interface Printable // Interface declaration
{
   void printDetails(); // Abstract method
}

class Student implements Printable // Implementing interface in Student
{
   String name;
   int roll;

   Student(String n, int r) // Constructor
   {
       name = n;
       roll = r;
   }
   public void printDetails() // Overriding interface method
   {
       System.out.println("Student Details -> Name: " + name + ", Roll
No: " + roll);
   }
```

```
}

class Employee implements Printable // Implementing interface in
Employee
{
   String empname;
   int empid;
   Employee(String n, int id) // Constructor
   {
       empname = n;
       empid = id;
   }
   public void printDetails() // Overriding interface method
   {
       System.out.println("Employee Details -> Name: " + empname + ",
ID: " + empid);
   }
}

public class InterfaceStudentEmployee
{
   public static void main(String args[]) // Main method
   {
       Scanner sc = new Scanner(System.in);
       System.out.print("Enter Student Name: ");
       String sn = sc.nextLine();
       System.out.print("Enter Student Roll No: ");
       int sr = sc.nextInt();
       sc.nextLine(); // Consuming newline
       System.out.print("Enter Employee Name: ");
       String en = sc.nextLine();
       System.out.print("Enter Employee ID: ");
       int ei = sc.nextInt();

       Printable p1 = new Student(sn, sr); // Interface reference to
Student object
       Printable p2 = new Employee(en, ei); // Interface reference to
Employee object

       System.out.println(" ");
       p1.printDetails(); // Calling overridden method for student
       p2.printDetails(); // Calling overridden method for employee
       sc.close();
   }
```

```
}
```

SwitchableInterface.java

```
// Assignment 11 - Exercise 2: Switchable Interface
import java.util.*;
interface Switchable // Interface declaration
{
   void turnOn(); // Abstract method
}

class Light implements Switchable // Implementing interface
{
   String room;
   Light(String r) // Constructor
   {
       room = r;
   }
   public void turnOn() // Overriding method
   {
       System.out.println("Light in " + room + " is now TURNED ON.");
   }
}

class Fan implements Switchable // Implementing interface
{
   int speed;
   Fan(int s) // Constructor
   {
       speed = s;
   }
   public void turnOn() // Overriding method
   {
       System.out.println("Fan is now TURNED ON at speed " + speed +
".");
   }
}

public class SwitchableInterface
{
   public static void main(String args[]) // Main method
   {
       Scanner sc = new Scanner(System.in);
       System.out.print("Enter Room for Light: ");
       String rm = sc.nextLine();
```

```
        System.out.print("Enter Fan Speed (1-5): ");
        int sp = sc.nextInt();

        Switchable s1 = new Light(rm); // Interface reference to Light
        Switchable s2 = new Fan(sp); // Interface reference to Fan

        System.out.println(" ");
        s1.turnOn(); // Executing Light method
        s2.turnOn(); // Executing Fan method
        sc.close();
    }
}
```

Output:

InterfaceStudentEmployee.java

```
Problems 10  Output  Debug Console  Terminal  Ports

cd "/Users/thegharkalaptop/Documents/Programming (The RS)/" && javac InterfaceStudentEmployee.java && java InterfaceStudentEmployee
thegharkalaptop@The-Ghar-Ka-Laptop Programming (The RS) % cd "/Users/thegharkalaptop/Documents/Programming (The RS)/" && javac InterfaceStudentEmployee.java && java InterfaceStudentEmployee
Enter Student Name: Raghav Sarda
Enter Student Roll No: 235
Enter Employee Name: Raghav Sarda
Enter Employee ID: 136

Student Details -> Name: Raghav Sarda, Roll No: 235
Employee Details -> Name: Raghav Sarda, ID: 136
thegharkalaptop@The-Ghar-Ka-Laptop Programming (The RS) %
```

SwitchableInterface.java

```
Problems 10  Output  Debug Console  Terminal  Ports

cd "/Users/thegharkalaptop/Documents/Programming (The RS)/" && javac SwitchableInterface.java && java SwitchableInterface
thegharkalaptop@The-Ghar-Ka-Laptop Programming (The RS) % cd "/Users/thegharkalaptop/Documents/Programming (The RS)/" && javac SwitchableInterface.java && java SwitchableInterface
Enter Room for Light: Bedroom
Enter Fan Speed (1-5): 3

Light in Bedroom is now TURNED ON.
Fan is now TURNED ON at speed 3.
thegharkalaptop@The-Ghar-Ka-Laptop Programming (The RS) %
```

Conclusion:

In this assignment, concepts of pure abstraction and multiple inheritance of type using interfaces were successfully implemented and verified. Interfaces provided a unified specification that distinct, unrelated classes like Student and Employee or Light and Fan adhered to, ensuring standardized method implementation and promoting high code modularity and loose coupling across the program.